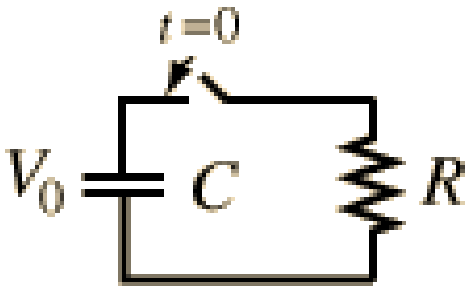
Pengisian & Pengosongan Kapasitor



$$V_C = \frac{Q}{C} = IR$$

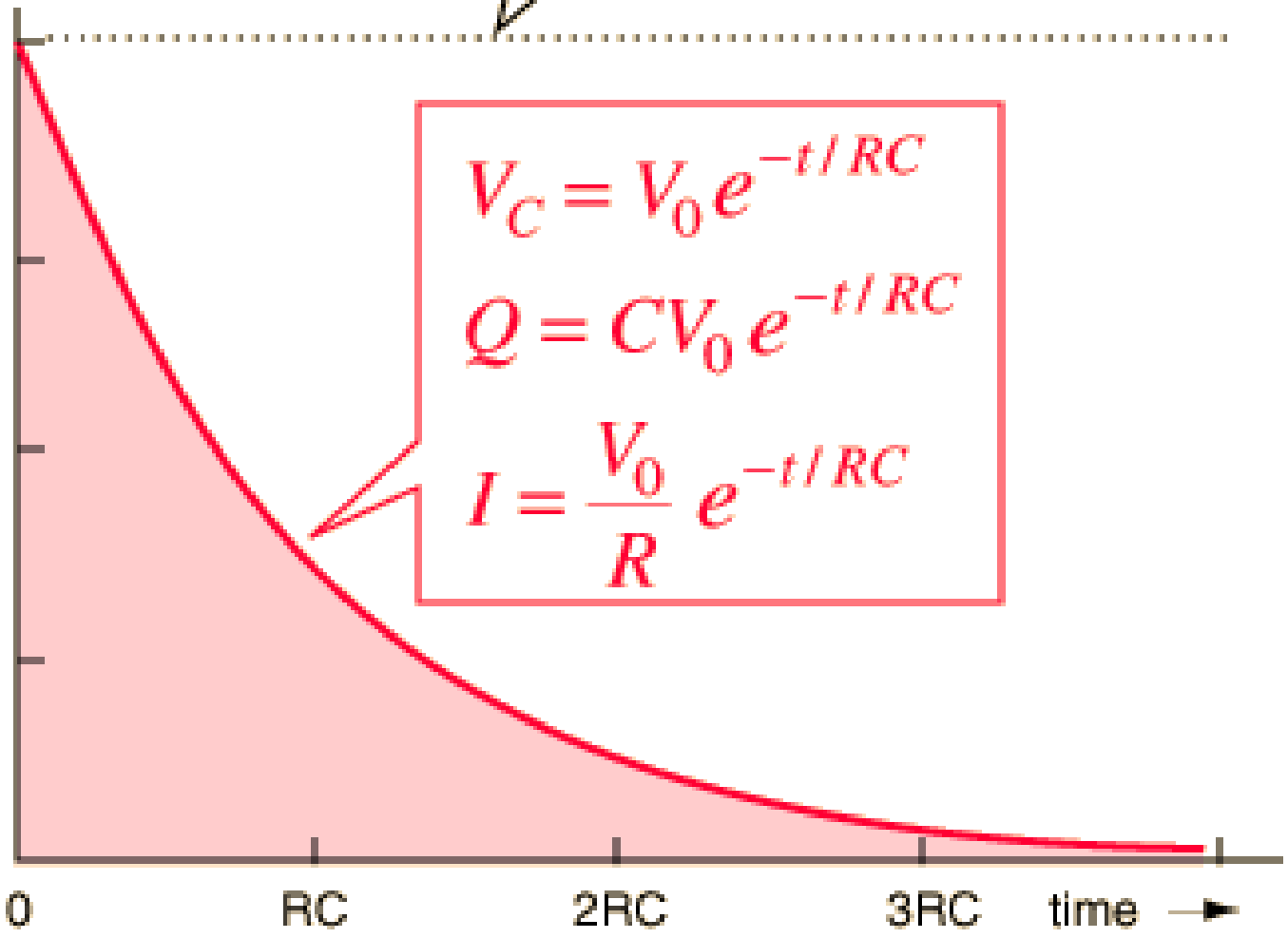
The voltage V_C, the current I, and the charge Q all follow the same type of decay curve when the switch is closed.

$$V_C = V_0,\ Q = CV_0,\ I = \frac{V_0}{R}$$

$$V_C = V_0 e^{-t/RC}$$

$$Q = CV_0 e^{-t/RC}$$

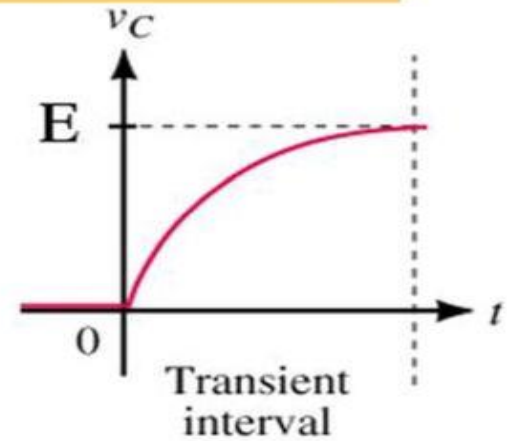
$$I = \frac{V_0}{R} e^{-t/RC}$$



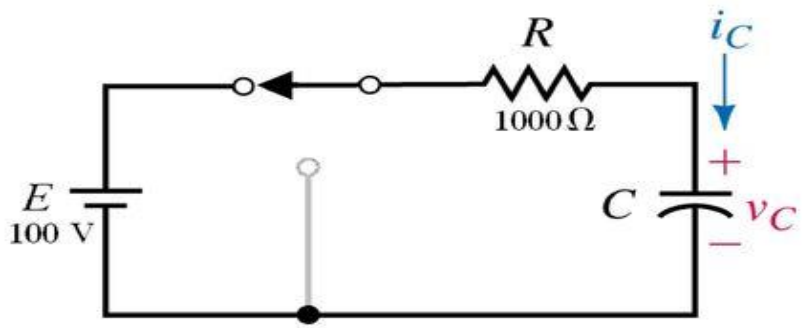
Capacitor Charging

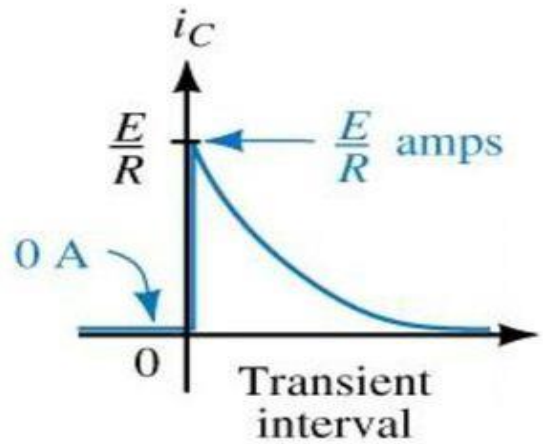
- As charge is stored in the capacitor, the voltage across the capacitor starts to rise.

$$v_C(t) = E(1 - e^{-t/RC})$$



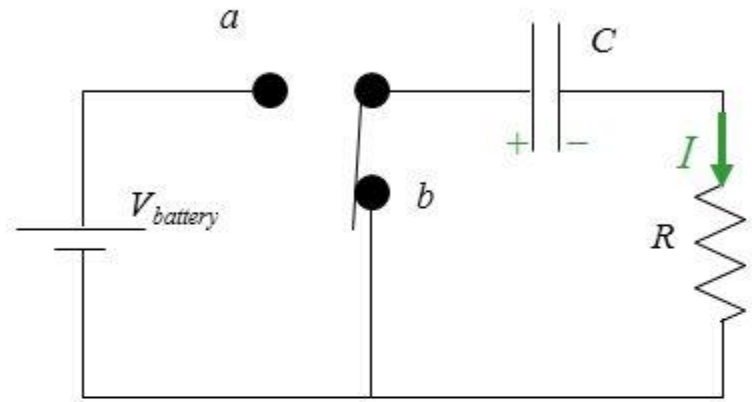
- This reduces the voltage across the resistor, therefore current in the circuit drops.





RC Circuit (Discharging)

Capacitor has $q_0 = CV$, Switch is moved to position "b"



Kirchoff's Voltage Rule

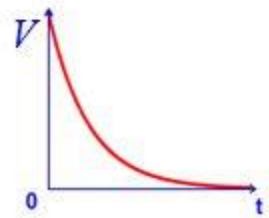
$$+\frac{q}{C} + IR = 0$$

Differential Equation

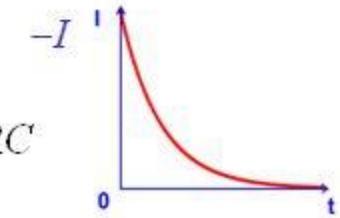
$$+\frac{q}{C} + \frac{dq}{dt}R = 0$$

$$-\frac{dt}{RC} = \frac{dq}{Q}$$

$$q(t) = q_0 e^{-t/RC}$$

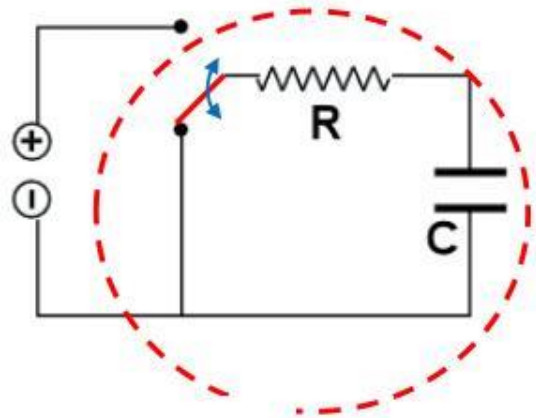


$$I(t) = I_0 e^{-t/RC}$$

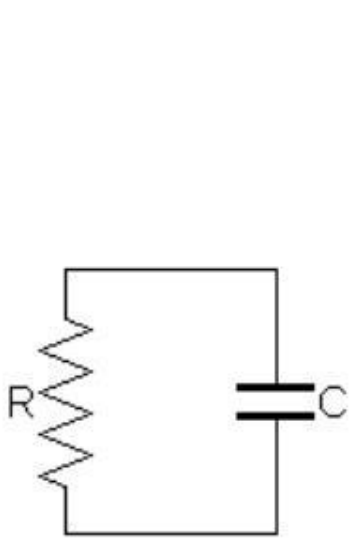


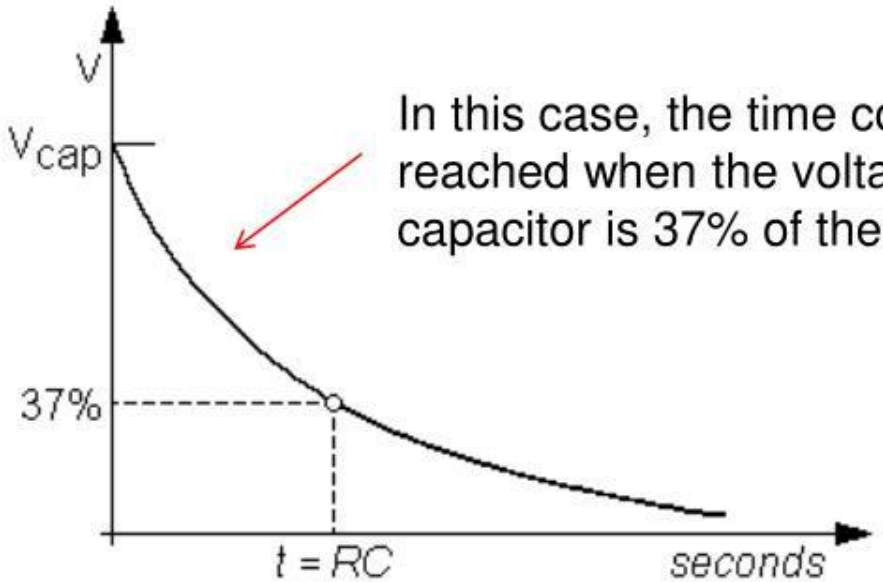
Capacitor Discharge – Capacitor's Voltage

The discharging graph for the capacitor is the same as that of the resistor. There WILL be a time delay due to the TIME CONSTANT of the circuit.



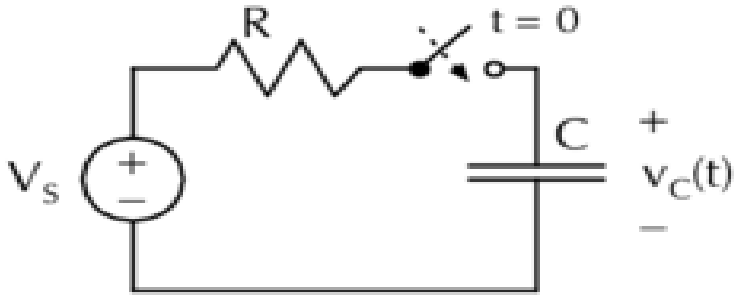
RC Discharge





In this case, the time constant is reached when the voltage of the capacitor is 37% of the EMF.

RC example 1



The switch in the simple *RC* circuit closes at $t = 0$. For the circuit, $V_S = 12$ V, $R = 10$ kΩ, C = 10 nF. The capacitor was initially charged to 6 volts. At what time does the capacitor voltage hit 10.5 V?

$V_{Ci} = 6$ V and $RC = 0.1$ ms $= 100$ μs.

$$v_C(t) = V_S - [V_S - V_{Ci}] \exp\left(-\frac{t}{RC}\right)$$

$$10.5V = 12V - [6V] \exp\left(-\frac{t'}{100\mu s}\right)$$

inverting it

$$t' = (100\mu s) \ln\left[\frac{6V}{12V - 10.5V}\right] = \boxed{130\mu s}$$

Example Problem 2b

The capacitor is now fully charged and at steady-state condition. The switch is opened to start the discharge cycle.

After the switch is open:(DISCHARGING)

a. determine how long it will take for the capacitor to fully discharge .

b. Identify the direction of current flow.

c. Write the equation for $v_c(t)$. Sketch the transient.

$$v_C(t) = Ee^{-t/RC}$$

